



Figure 1: **GCVaR vs. policy gradient.** (A,B) Average return (A) and CVaR ($\alpha = 0.05$) of the return (B) for CVaRSGD and standard policy-gradient vs. iteration. (C) Histogram (counts from 10,000 independent runs) of the total return of the final policies. The lower plot is a zoom-in on the left-tail, and clearly shows the risk-averse behavior of the CVaRSGD policy. (D) Final policy parameters. Note the difference in the Board Well feature, which encourages risk taking. (E) CVaR ($\alpha = 0.01$) of the return for CVaRSGD vs. iteration, with and without importance sampling.

tions strengthened the difference between the risk-sensitive and nominal policies, as they induce a tradeoff between clearing many ‘single’ lines with a low profit, or waiting for the more profitable, but less frequent, ‘batches’.

We used the softmax policy, with the feature set of ? (?). Starting from a fixed policy parameter θ_0 , which was obtained by running several iterations of standard policy gradient (giving both methods a ‘warm start’), we ran both CVaRSGD and standard policy gradient⁴ for enough iterations such that both algorithms (approximately) converged. We set $\alpha = 0.05$ and $N = 1000$.

In Fig. 1A and Fig. 1B we present the average return $V(\theta)$ and CVaR of the return $J(\theta)$ for the policies of both algorithms at each iteration (evaluated by MC on independent trajectories). Observe that for CVaRSGD, the average return has been compromised for a higher CVaR value.

This compromise is further explained in Fig. 1C, where we display the reward distribution of the final policies. It may be observed that the left-tail distribution of the CVaR policy is significantly lower than the standard policy. For the risk-sensitive decision maker, such results are very important, especially if the left-tail contains catastrophic outcomes, as is common in many real-world domains, such as finance. To better understand the differences between the policies, we compare the final policy parameters θ in Fig. 1D. The most significant difference is in the parameter that corresponds to the Board Well feature. A *well* is a succession of unoccupied cells in a column, such that their left and right cells are both occupied. The controller trained by CVaRSGD has a smaller negative weight for this feature, compared to the standard controller, indicating that actions which create deep-wells are repressed. Such wells may lead to a high reward when they get filled, but are risky as they heighten the board.

To demonstrate the importance of IS in optimizing the CVaR when α is small, we chose $\alpha = 0.01$, and $N = 200$, and compared CVaRSGD against its IS version, IS_CVaRSGD,

⁴Standard policy gradient is similar to CVaRSGD when $\alpha = 1$. However, it is common to subtract a baseline from the reward in order to reduce the variance of the gradient estimate. In our experiments, we used the average return $\langle r \rangle$ as a baseline, and our gradient estimate was $\frac{1}{N} \sum_{i=1}^N \frac{\partial \log f_Y(y_i; \theta)}{\partial \theta_j} (r(x_i, y_i) - \langle r \rangle)$.

described in the supplementary material. As Fig. 1E shows, IS_GCVaRSGD converged significantly faster, improving the convergence rate by more than a factor of 2. The full details are provided in the supplementary material.

6 Conclusion and Future Work

We presented a novel LR-style formula for the gradient of the CVaR performance criterion. Based on this formula, we proposed a sampling-based gradient estimator, and a stochastic gradient descent procedure for CVaR optimization that is guaranteed to converge to a local optimum. To our knowledge, this is the first extension of the LR method to the CVaR performance criterion, and our results extend CVaR optimization to new domains.

We evaluated our approach empirically in an RL domain: learning a risk-sensitive policy for Tetris. To our knowledge, such a domain is beyond the reach of existing CVaR optimization approaches. Moreover, our empirical results show that optimizing the CVaR indeed results in useful risk-sensitive policies, and motivates the use of simulation-based optimization for risk-sensitive decision making.

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